

A moving boundary model for food isothermal drying and shrinkage: One-dimensional versus two-dimensional approaches

Alessandra Adrover¹  | Antonio Brasiello² 

¹Dipartimento di Ingegneria Chimica, Materiali e Ambiente, Sapienza Università degli Studi di Roma, Rome, Italy

²Dipartimento di Ingegneria Industriale, Università degli Studi di Salerno, Fisciano (SA), Italy

Correspondence

Alessandra Adrover, Dipartimento di Ingegneria Chimica, Materiali e Ambiente, Sapienza Università degli Studi di Roma, via Eudossiana 18, Rome 00184, Italy.
Email: alessandra.adrover@uniroma1.it

Abstract

We investigate and quantify the error in the estimate of water diffusivity resulting from the adoption of one-dimensional (1-d) models to describe the drying process of thin discoidal or long-thin cylindrical samples. Numerical results obtained with the 1-d and the 2-d moving boundary models recently proposed by Adrover, Brasiello, and Ponso (2019a, 2019b) are compared for different sample aspect ratios. The overestimate error of 1-d model predictions of diffusivity falls below 5% for an aspect ratio radius-over-thickness $R_0/L_0 > 10$ for discoidal samples while, for cylindrical samples, the aspect ratio height-to-radius L_0/R_0 must be greater than 15 to get a similar level of accuracy. To overcome this issue we propose a modified 1-d model able to take into account water flux from lateral surfaces (usually neglected in a 1-d approach) thus reducing the overestimate error of water diffusivity below 2% even for low-intermediate values of the aspect ratio. The 1-d modified model is successfully applied to experimental data of convective drying of eggplant and chayote discoidal samples.

Practical applications

End consumer is increasingly demanding for ever more standardized and high quality food products. To meet these requirements, food industries are challenged to refine process design and control tools in order to be able to take into account the variability of both food matrices and operating conditions. This objective cannot be reached without the help of mathematical models based on first principles as the one here discussed and applied to food drying. The comparison between one- and two-dimensional approaches and the resulting quantification of errors in the adoption of simpler one-dimensional models have a great practical interest for industrial applications as they provide reliable criteria to obtain good predictions even when the whole set of the needed experimental data are not available or too expensive. The developed method can be easily adopted to predict final food quality (moisture content, sample volume reduction, and deformation) and therefore to implement more refined control systems and to optimize process-operating conditions, thus saving experimental and processing costs.

1 | INTRODUCTION

Since few years, mathematical models started to be applied in food science and engineering not only to better understand all complex physical phenomena involved (Brasiello, Crescitelli, & Milano, 2012; Brasiello, Russo, Siettos, Milano, & Crescitelli, 2010; Poschet, Geeraerd, Scheerlinck, Nicolai, & Van Impe, 2003) but also to improve methods and tools of process design and control (Li & Mason, 2002; Liu & Bakker-Arkema, 1997; Terrazas-Velarde, Peglow, & Tsotsas, 2009) as well as scale-up strategies from laboratory or pilot-plant to larger industrial scales.

Food industries are challenged to develop ever better ways to predict food processes outputs and to obtain higher quality products by varying operating conditions. This is possible through mathematical modeling based on physico-chemical principles through which the desired information are achieved as well as deeper insights into some important aspects of food technology (e.g., microbiology (Poschet et al., 2003), phase transitions (Pizzirusso, Brasiello, De Nicola, Marangoni, & Milano, 2015), packaging (Ansari & Datta, 2003), structural analysis (Madiouli, Sghaier, Lecomte, & Sammouda, 2012; Nowicki, Davis, & Scriven, 1992; Wang & Brennan, 1995; Xu, Mujumdar, & Yu, 2008)).

In food science, theories behind models, suitable for description of specific phenomena occurring at different time and length scales, are often borrowed from other research fields such as molecular dynamics (Frenkel & Smit, 1996; Nivelle, Beghin, Bosmans, & Delcour, 2018), computational chemistry, stochastic dynamics (Giona, Brasiello, & Crescitelli, 2017a, 2017b; Helmroth, Varekamp, & Dekker, 2005), Monte Carlo methods (Defraeye, 2014), nonequilibrium mechanics and thermodynamics (Bird, Stewart, & Lightfoot, 2002; Datta, 2007). The synergistic use of different approaches can bridge together different spatio-temporal scales (Brasiello, Crescitelli, & Giona, 2016; Defraeye, 2014; Giona, Brasiello, & Crescitelli, 2015).

Food dehydration often involves deep modifications in sample global aspect, shape, porosity, and bulk density (Ramos, Brandao, & Silva, 2003). Significant volume reduction and deformation connected to deep structural modifications (namely shrinkage) observed during drying (Aprajeeta, Gopirajah, & Anandharamakrishnan, 2015; Ortiz-Garcia-Carrasco et al., 2015; Pacheco-Aguirre et al., 2015) make the problem very difficult to approach from the theoretical/modeling point of view. This is the reason why the consolidated approach consists in the use of simple semi-heuristic kinetic models (Mayor & Sereno, 2004). However, mathematical models based on physico-chemical principles could be useful tools in food drying modeling in order to predict the evolution of the state variables (namely the water concentration) and volume changes (Curcio & Aversa, 2014; Vu & Tsotsas, 2018).

One of the simplest model proposed, able to predict water content evolution and to give a reasonable estimate of volume reduction, is that developed in Brasiello et al. (2013, 2017). The model is based on Fickian transport theory (Crank, 1975) and the structural changes are given through the introduction of a dependence of the diffusion coefficient on water concentration. Despite the good results in terms

of prediction of moisture content as well as of water spatial profiles, the model is not able to predict volume evolution in time.

A significant improvement in this theoretical line is represented by the moving-boundary model recently proposed by Adrover, Brasiello, and Ponso (2019a, 2019b); Adrover and Brasiello (2019) inspired by the seminal works by Tu and Ouano (1977) and Papanu, Soane (Soong, Bell, and Hess (1989) on solvent transport and swelling of polymeric matrices.

The core of the model is the introduction of a pointwise shrinkage velocity assumed proportional to the water concentration gradient times a proportionality shrinkage function α tuning, at each point of the system, the relationship between water flux and volume reduction. As a matter of fact, the shrinkage function α represents the constitutive equation –the fingerprint– of the food material under investigation. It can be determined from experimental “calibration curves” relating the moisture fraction X/X_0 to the rescaled volume V/V_0 during drying. In previous works, Adrover, et al. (2019a, 2019b) widely discussed about this point and proposed two different approaches, a fully analytic and a shortcut numerical method, to determine the shrinkage function from calibration curves X/X_0 versus V/V_0 . The moving-boundary model is able to describe water profiles, sample volume reduction and surface deformation during drying with great accuracy.

In many practical cases, while time evolution of moisture content is easily measured by thermogravimetric analysis, all measurements that are required for sample volume monitoring during drying are not often recorded due to the irregular shape of the sample or for intrinsic limitations of experimental techniques, for example, optical methods, adopted to follow the sample shape evolution. In many cases only data related to one main shrinkage direction are available, for example, radius for long thin cylinders or thickness for thin slab or discoidal samples in these cases.

In order to successfully apply the moving boundary model we can adopt two different approaches: (a) we can estimate sample volume from thickness or radius measurements, as discussed in Section 6, and therefore apply a two- or three-dimensional model; (b) we can apply a one-dimensional (1-d) model thus neglecting water flux and sample shrinkage along the non-monitored directions. For example, for a discoidal sample, shrinkage along the radial direction is neglected as well as water flux from the lateral cylindrical surface of the discoid.

The main drawback in the adoption of a simplified 1-d model resides in an unavoidable overestimation of transport parameters, mainly the water diffusion coefficient D .

In this article, the comparison between the 1-d approximation and the 2-d model for both discoidal and cylindrical samples is carried out. The aim is to quantify, in two simple cases, the error in the diffusion coefficient estimate when the 1-d approximation is adopted and to identify the minimum sample aspect ratio (thickness-over-radius for discoidal samples and height-over-radius for long cylindrical samples) that can be adopted in order to reduce the overestimation error for D below 5%.

A novel approach to reduce errors in diffusion coefficient estimation due to one dimensional approximation is also proposed. The

procedure is demonstrated to be effective on two literature cases, specifically eggplant drying (Brasiello et al., 2013) and chayote drying (Ruiz-López, Ruiz-Espinosa, Arellanes-Lozada, Bárcenas-Pozos, & García-Alvarado, 2012), the latter being characterized by a highly nonlinear calibration curve. Numerical and computational issues about 1-d and 2-d implementation of the moving boundary model are also discussed in detail.

2 | 1-D AND 2-D MOVING BOUNDARY MODELS

We briefly recall the model equations derived in Adrover et al. (2019b). The transport equations describing the space-time evolution of the water volume fraction $\phi(\mathbf{x}, t)$ and of the sample boundary $S(t)$ are

$$\frac{\partial \phi}{\partial t} = \nabla \cdot (D \nabla \phi (1 - \alpha \phi)), \quad \mathbf{x} \in V(t) \quad (1)$$

$$-D \nabla \phi \cdot \mathbf{n}|_{\mathbf{x}_b} = h_m K_{eq} (\rho_{air} / \rho_s) (\phi|_{\mathbf{x}_b} - \phi_{eq}), \quad \mathbf{x}_b \in S(t) \quad (2)$$

$$\frac{d\mathbf{x}_b}{dt} = \mathbf{v}_s = \alpha D \nabla \phi|_{\mathbf{x}_b}, \quad \mathbf{x}_b \in S(t) \quad (3)$$

$$\phi(\mathbf{x}, 0) = \phi_0, \quad S(0) = S_0, \quad V(0) = V_0 \quad (4)$$

where h_m is a mass transfer coefficient, ρ_{air} is the air density at the operating temperature, ρ_s is the solid (pulp) density and K_{eq} is the water partition ratio between the gas and the solid phases $H = K_{eq}(\rho_w / \rho_s)\phi$, H being the absolute air humidity (kg water/kg dry air).

The shrinkage velocity $\mathbf{v} = -\alpha(\phi) \mathbf{J} = \alpha(\phi) D \nabla \phi$ is proportional to the water concentration gradient times a proportionality function $\alpha(\phi)$ tuning, at each point of the system, the relationship between water flux $\mathbf{J}$ and volume reduction. The shrinkage velocity, $\mathbf{v}_s$, evaluated at the boundary of the physical domain Equation (3), controls the movement of the boundary. The water diffusivity D is assumed constant in space and time.

By introducing the dimensionless space and time variables $\tau = tD/L_r^2$, $\tilde{\mathbf{x}} = \mathbf{x}/L_r$, $\tilde{V} = V/L_r^3$, $\tilde{S} = S/L_r^2$, and the dimensionless differential operators $\tilde{\nabla} = \nabla/L_r$, $\tilde{\nabla} \cdot = \frac{\nabla \cdot}{L_r}$, L_r being a characteristic reference length, the moving boundary model equations for the normalized water volume fraction $\psi = \phi/\phi_0$ attain the form

$$\frac{\partial \psi}{\partial \tau} = \tilde{\nabla} \cdot (\tilde{\nabla} \psi (1 - \alpha(\psi) \phi_0 \psi)), \quad \tilde{\mathbf{x}} \in \tilde{V}(\tau) \quad (5)$$

$$-\tilde{\nabla} \psi \cdot \mathbf{n}|_{\tilde{\mathbf{x}}_b} = Bi_m (\psi|_{\tilde{\mathbf{x}}_b} - \psi_{eq}), \quad \tilde{\mathbf{x}}_b \in \tilde{S}(t) \quad (6)$$

$$\frac{d\tilde{\mathbf{x}}_b}{d\tau} = \alpha(\psi) \phi_0 \tilde{\nabla} \psi|_{\tilde{\mathbf{x}}_b}, \quad \tilde{\mathbf{x}}_b \in \tilde{S}(\tau) \quad (7)$$

where Bi_m in the mass transfer Biot number $Bi_m = \frac{h_m L_r}{D} K_{eq} \frac{\rho_{air}}{\rho_s}$. Initial conditions are $\tilde{V}(0) = \tilde{V}_0$, $\tilde{S}(0) = \tilde{S}_0$ and $\psi(\tilde{\mathbf{x}}, 0) = 1$, assuming initial uniform water distribution.

For a 2-d axial symmetric cylindrical sample $-L_0/2 \leq z \leq L_0/2$, $0 \leq r \leq R_0$ we introduce the dimensionless space variables $(\rho, \zeta) = (r/L_r, z/L_r)$ and the dimensionless differential operators

$$\nabla f = \frac{\partial f}{\partial \rho} \mathbf{e}_\rho + \frac{\partial f}{\partial \zeta} \mathbf{e}_\zeta, \quad \nabla \cdot \mathbf{f} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho f_\rho) + \frac{\partial f_\zeta}{\partial \zeta} \quad (8)$$

Two different cases will be analyzed: the discoidal sample and the long-thin cylinder.

2.1 | The discoidal sample

In dealing with a discoidal sample with $R_0 > L_0$, we choose as a reference length $L_r = L_0$ so that the dimensionless initial domain is $(\rho, \zeta) \in [0, R_0/L_0] \times [-1/2, 1/2]$, R_0/L_0 being the discoidal aspect ratio.

If we adopt a 2-d model for describing the drying process of a discoidal sample, the shrinkage factor $\alpha(\psi)$ must be estimated from experimental data of rescaled volume V/V_0 versus free moisture fraction X/X_0 ("calibration curve") by setting $\alpha(\psi) = \alpha_V(\psi_b)$ where $\alpha_V(\psi_b)$ is evaluated from the integral balance equation

$$\frac{d(X/X_0)}{dt} = \frac{1}{\phi_0 \alpha_V(\psi_b)} \frac{d(V/V_0)}{dt} \rightarrow \alpha_V(\psi_b) \simeq \frac{1}{\phi_0} \frac{dV/V_0}{dX/X_0} = \frac{G_V(g(\psi_b, Bi_m))}{\phi_0} \quad (9)$$

where

$$\frac{V}{V_0} = G_V\left(\frac{X}{X_0}\right), \quad G_V\left(\frac{X}{X_0}\right) = \frac{dG_V}{d(X/X_0)}, \quad \left(\frac{X}{X_0}\right) = g(\psi_b, Bi_m) \quad (10)$$

and ψ_b is the rescaled water volume fraction evaluated at the probe point initially placed at $(\rho, \zeta) = (0, 1/2)$; for a discussion about the choice of the probe point see Adrover, et al. (2019a, 2019b).

For high values of the aspect ratio R_0/L_0 a 1-d model can be readily adopted, describing the time evolution of the rescaled water volume fraction $\psi(\zeta, \tau)$ along the axial coordinate (associated with the smallest initial dimension L_0) and the time evolution of the sample thickness $\tilde{L}(\tau)$ (uniform along the radial direction)

$$\frac{\partial \psi}{\partial \tau} = \frac{\partial}{\partial \zeta} \left(\frac{\partial \psi}{\partial \zeta} (1 - \alpha(\psi) \phi_0 \psi) \right), \quad \zeta \in \left(-\frac{\tilde{L}(\tau)}{2}, \frac{\tilde{L}(\tau)}{2} \right) \quad (11)$$

$$-\frac{\partial \psi}{\partial \zeta} \Big|_{\zeta = \tilde{L}/2} = Bi_m (\psi|_{\tilde{L}/2} - \psi_{eq}) \quad (12)$$

$$\frac{\partial \psi}{\partial \zeta} \Big|_{\zeta = -\tilde{L}/2} = Bi_m (\psi|_{-\tilde{L}/2} - \psi_{eq}) \quad \text{or equivalently} \quad \frac{\partial \psi}{\partial \zeta} \Big|_{\zeta = 0} = 0 \quad (13)$$

$$\frac{d(\tilde{L}/2)}{d\tau} = \alpha(\psi) \phi_0 \frac{\partial \psi}{\partial \zeta} \Big|_{\tilde{L}/2}, \quad \frac{\tilde{L}(0)}{2} = \frac{1}{2} \quad (14)$$

In this way one neglects the radial shrinkage as well as the water flux from the discoidal lateral surface.

If we adopt a 1-d model for describing the drying process of a discoidal sample, the shrinkage factor $\alpha(\psi)$ must be estimated from experimental data of rescaled thickness L/L_0 versus free moisture fraction X/X_0 as $\alpha(\psi) = \alpha_L(\psi_b)$ where

$$\frac{d(X/X_0)}{d\tau} = \frac{1}{\phi_0 \alpha_L(\psi_b)} \frac{d(L/L_0)}{d\tau} \rightarrow \alpha_L(\psi_b) \simeq \frac{1}{\phi_0} \frac{dL/L_0}{dX/X_0} = \frac{G_L(g(\psi_b, Bi_m))}{\phi_0} \quad (15)$$

where

$$\frac{L}{L_0} = G_L\left(\frac{X}{X_0}\right), \quad G_L\left(\frac{X}{X_0}\right) = \frac{dG_L}{d(X/X_0)}, \quad \left(\frac{X}{X_0}\right) = g(\psi_b, Bi_m) \quad (16)$$

and ψ_b is the rescaled water volume fraction evaluated at the probe point initially placed at $\zeta = (1/2)$, i.e., $\psi_b = \psi(\tilde{L}/2, \tau)$.

If both experimental calibration curves V/V_0 versus X/X_0 and L/L_0 versus X/X_0 are available one can apply in a straightforward way both the 2-d and the 1-d models, although a 2-d model has to be preferred since the free moisture fraction X/X_0 accounts for water flux from the entire surface sample, including the lateral surface.

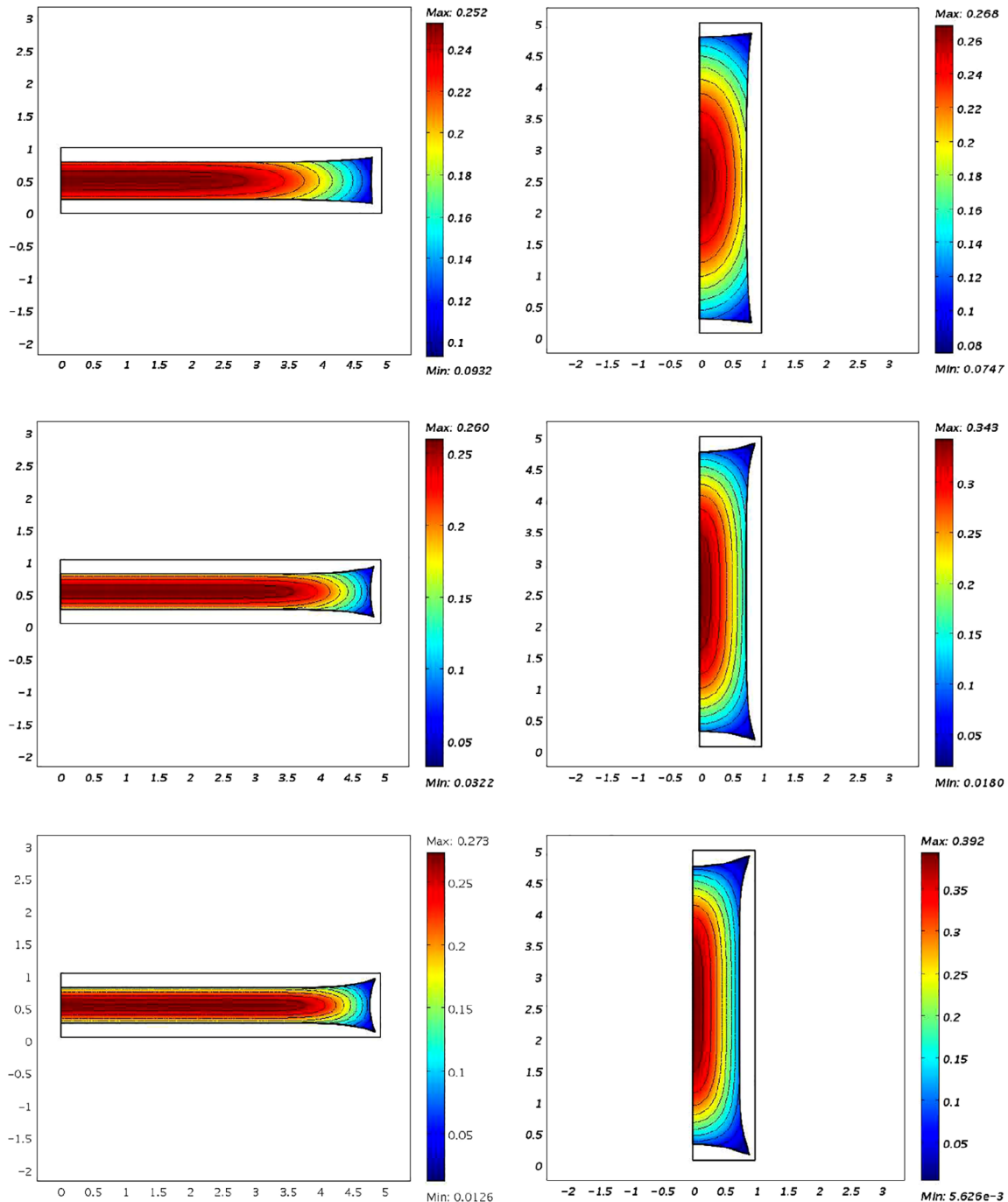


FIGURE 1 Contour plot of rescaled water volume fraction $\psi(\rho, \zeta)$ for $X/X_0 = 0.1$ and for increasing values of $Bi_m = 1, 3, 5$ (from top to bottom). Left panel: discoidal sample ($R_0/L_0 = 5$); Right panel: cylindrical sample ($L_0/R_0 = 5$)

If only experimental data for L/L_0 versus X/X_0 are available, one can apply the 2-d model by preliminary estimating, from L/L_0 data, the actual rescaled volume V/V_0 . We discuss in detail this issue in Section 6 and in Sections 7 and 8 when dealing with literature experimental data on convective drying of eggplants and chayote discoidal samples, respectively.

2.2 | The long-thin cylinder

In dealing with a cylindrical sample with $L_0 > R_0$ we choose as a reference length $L_r = R_0$ so that the dimensionless initial domain is $(\rho, \zeta) \in [0, 1] \times [-L_0/(2R_0), L_0/(2R_0)]$, L_0/R_0 being the cylinder aspect ratio.

As in the case of a discoidal sample, the shrinkage factor can be evaluated from experimental data V/V_0 versus X/X_0 by making use of Equations (9) and (10) but the probe point should be conveniently identified with the point with initial position $(\rho, \zeta) = (1, 0)$ as the sample shrinkage occurs mainly along the radial direction.

For long-thin cylinders, that is, for an high aspect ratio L_0/R_0 , one can adopt a 1-d model describing the time evolution of the rescaled water volume fraction $\psi(\rho, \tau)$ along the radial coordinate ρ and the radial shrinkage $\tilde{R}(\tau)$, thus neglecting the axial shrinkage and the water flux from the top and bottom cylinder surfaces. The corresponding 1-d model reads as

$$\frac{\partial \psi}{\partial \tau} = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \psi}{\partial \rho} (1 - \alpha(\psi) \phi_0 \psi) \right), \quad \rho \in (0, \tilde{R}(\tau)) \quad (17)$$

$$\left. \frac{\partial \psi}{\partial \rho} \right|_{\rho=0} = 0, \quad -\left. \frac{\partial \psi}{\partial \rho} \right|_{\rho=\tilde{R}} = Bi_m (\psi|_{\tilde{R}} - \psi_{eq}) \quad (18)$$

$$\frac{d\tilde{R}}{d\tau} = \alpha(\psi) \phi_0 \left. \frac{\partial \psi}{\partial \rho} \right|_{\rho=\tilde{R}}, \quad \tilde{R}(0) = 1 \quad (19)$$

The shrinkage factor $\alpha(\psi)$ can be estimated from experimental data of rescaled radius R/L_0 versus free moisture fraction X/X_0 by adopting Equations (9) and (10) where $V/V_0 = (R/R_0)^2$ (since axial

shrinkage is neglected) with the probe point initially placed at $\rho = 1$, that is, $\psi_b = \psi(\tilde{R}, \tau)$.

Like in the case of a discoidal sample, if both calibration curves V/V_0 versus X/X_0 and R/R_0 versus X/X_0 are available one can apply in a straightforward way both the 2-d and the 1-d models, although a 2-d model has to be preferred.

If only experimental data for R/R_0 versus X/X_0 are available, one can apply the 2-d model by preliminary estimating, from R/R_0 data, the actual rescaled volume V/V_0 . A simple quantitative way for relating V/V_0 to R/R_0 is presented in Section 6.

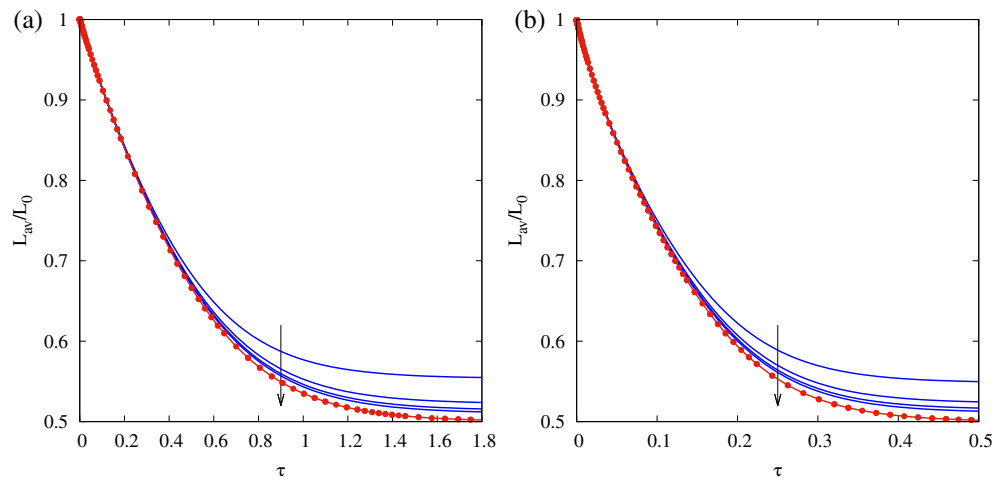
3 | COMPUTATIONAL ISSUES

The 1-d and the 2-d moving boundary problems have been numerically solved using finite elements method (FEM) in Comsol Multiphysics 3.5. The convection-diffusion package (conservative form) has been coupled with ALE (Arbitrary Lagrangian Eulerian) moving mesh, allowing re-meshing during the time evolution of the physical domain. Free displacement induced by boundary velocity conditions has been set. Lagrangian quadratic elements have been chosen. The linear solver adopted is UMFPACK, with relative tolerance 10^{-3} and absolute tolerance 10^{-6} . The number of finite elements is 10^4 for 1-d simulations and $2 \times 10^5 - 5 \times 10^5$ for 2-d simulations (increasing with the aspect ratio) with a non-uniform mesh. Smaller elements (maximum element size 5×10^{-3} dimensionless units) have been located close to the moving boundaries in order to accurately compute concentration gradients controlling the velocity of the moving fronts.

Focusing on computational issues, the real advantage of adopting a 1-d model is that the 1-d moving boundary problem can be easily transformed into a transport problem with fixed boundaries, including the boundary dynamics into the transport equations.

As an example, let us consider the 1-d model with radial shrinkage Equations (17)–(19). By introducing the dimensionless variable $\tilde{\rho} = \rho/\tilde{R}(\tau)$ we can rewrite the model equations Equations (17)–(19) in terms of the new rescaled water volume fraction $\tilde{\psi}(\tilde{\rho}, \tau) = \psi(\rho, \tau)$ defined on a fixed domain $\tilde{\rho} \in [0, 1]$.

FIGURE 2 L_{av}/L_0 versus $\tau = tD/L_0^2$ for a discoidal sample with $R_0 > L_0$ for increasing values of the aspect ratio $R_0/L_0 = 2.5, 5, 7.5, 10$. Arrow indicates increasing values of R_0/L_0 . Comparison between the 1-d model (insensitive to the aspect ratio, red dots) and 2-d model (continuous lines) for $\alpha(\psi) = C = 1$, $\phi_0 = 0.5$. (a) $Bi_m = 1$; (b) $Bi_m = 5$



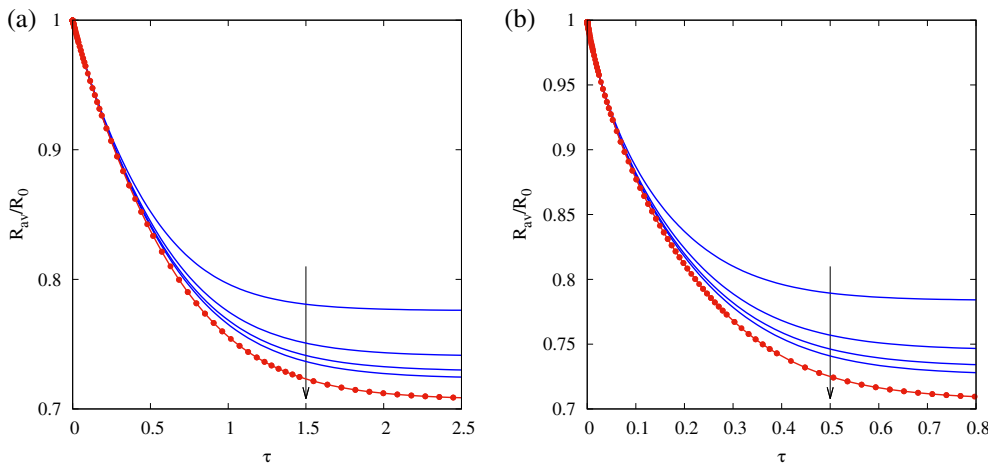


FIGURE 3 R_{av}/R_0 versus $\tau = tD/R_0^2$ for a cylindrical sample with $L_0 > R_0$ for increasing values of the aspect ratio $L_0/R_0 = 2.5, 5, 7.5, 10$. Arrow indicates increasing values of L_0/R_0 . Comparison between the 1-d model (insensitive to the aspect ratio, red dots) and 2-d model (continuous lines) for $\alpha(\psi) = C = 1$, $\phi_0 = 0.5$. (a) $Bi_m = 1$; (b) $Bi_m = 5$

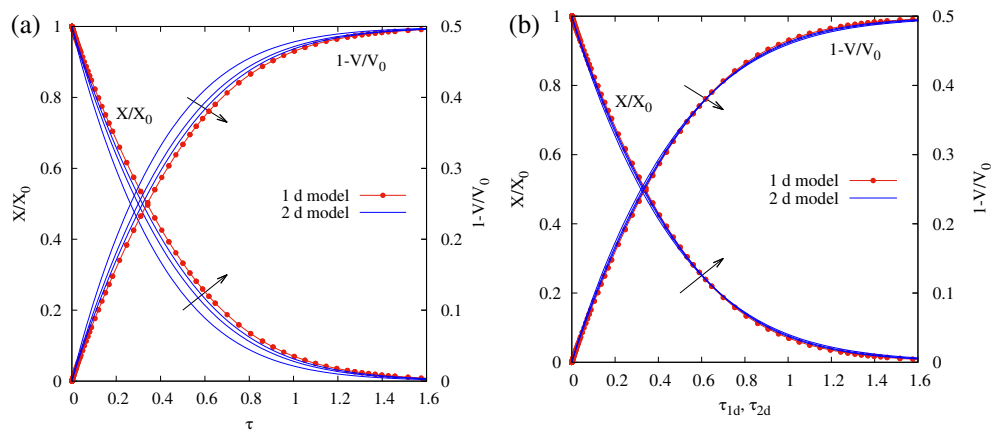


FIGURE 4 (a) X/X_0 and $1 - V/V_0$ versus $\tau = tD/L_0^2$ for a cylindrical slab with $R_0 > L_0$ (discoidal sample) for increasing values of the aspect ratio $R_0/L_0 = 2.5, 5, 10$. Arrows indicate increasing values of R_0/L_0 . Comparison between the 1-d model (insensitive to the aspect ratio, red dots) and 2-d model (continuous lines) for $\alpha(\psi) = C = 1$, $\phi_0 = 0.5$, $Bi_m = 1$; (b) X/X_0 and $1 - V/V_0$ versus $\tau_{2d} = tD_{2d}/L_0^2$ for the 2-d model with $R_0/L_0 = 2.5, 5, 10$ and $D_{2d} = D_{1d}/\beta(R_0/L_0)$ collapsing onto the corresponding curves X/X_0 and $1 - V/V_0$ versus $\tau_{1d} = tD_{1d}/L_0^2$ for $\beta(2.5) = 1.21$, $\beta(5) = 1.11$, and $\beta(10) = 1.05$

By making use of the following identities

$$\frac{\partial \psi}{\partial \tau} = \frac{\partial \tilde{\psi}}{\partial \tau} + \frac{\partial \tilde{\psi}}{\partial \tilde{\rho}} \frac{\partial \tilde{\rho}}{\partial \tau} = \frac{\partial \tilde{\psi}}{\partial \tau} - \left(\frac{\tilde{\rho}}{\tilde{R}} \frac{d\tilde{R}}{d\tau} \right) \frac{\partial \tilde{\psi}}{\partial \tilde{\rho}} \quad (20)$$

$$\frac{\partial \psi}{\partial \rho} = \frac{\partial \tilde{\psi}}{\partial \tilde{\rho}} \frac{\partial \tilde{\rho}}{\partial \rho} = \frac{1}{\tilde{R}} \frac{\partial \tilde{\psi}}{\partial \tilde{\rho}} \quad (21)$$

we obtain the transport model for $\tilde{\psi}(\tilde{\rho}, \tau)$ on a fixed domain $\tilde{\rho} \in [0, 1]$

$$\frac{\partial \tilde{\psi}}{\partial \tau} = \left(\frac{\tilde{\rho}}{\tilde{R}} \frac{d\tilde{R}}{d\tau} \right) \frac{\partial \tilde{\psi}}{\partial \tilde{\rho}} + \frac{1}{\tilde{R}^2} \frac{\partial}{\partial \tilde{\rho}} \left(\tilde{\rho} \frac{\partial \tilde{\psi}}{\partial \tilde{\rho}} (1 - \alpha(\tilde{\psi}) \phi_0 \tilde{\psi}) \right), \quad \tilde{\rho} \in (0, 1) \quad (22)$$

$$\frac{\partial \tilde{\psi}}{\partial \tilde{\rho}} \Big|_{\tilde{\rho}=0} = 0, \quad -\frac{\partial \tilde{\psi}}{\partial \tilde{\rho}} \Big|_{\tilde{\rho}=1} = \tilde{R} Bi_m \left(\tilde{\psi} \Big|_{\tilde{\rho}=1} - \psi_{eq} \right) \quad (23)$$

coupled with the ordinary differential equation describing the time-evolution of $\tilde{R}(\tau)$ in terms of $\tilde{\psi}$

$$\frac{d\tilde{R}}{d\tau} = \frac{1}{\tilde{R}} \left(\alpha(\tilde{\psi}) \phi_0 \frac{\partial \tilde{\psi}}{\partial \tilde{\rho}} \right) \Big|_{\tilde{\rho}=1}, \quad \tilde{R}(0) = 1 \quad (24)$$

From the numerical solution of the fixed boundary transport model Equations (22)–(24) the free moisture fraction X/X_0 and the rescaled volume V/V_0 can be evaluated at each time instant τ as

$$\frac{X}{X_0} = \int_0^{\tilde{R}(\tau)} \psi(\rho, \tau) 2\pi \rho d\rho = \tilde{R}^2 \int_0^1 \tilde{\psi}(\tilde{\rho}, \tau) 2\pi \tilde{\rho} d\tilde{\rho}, \quad \frac{V}{V_0} = \tilde{R}^2 \quad (25)$$

4 | LIMITS OF APPLICABILITY OF 1-D MODELS

1-d models can be reasonably applied when the initial physical domain is characterized by an high aspect ratio $R_0/L_0 > 1$ for discoidal samples and $L_0/R_0 > 1$ for cylindrical samples. In both cases, by applying a 1-d

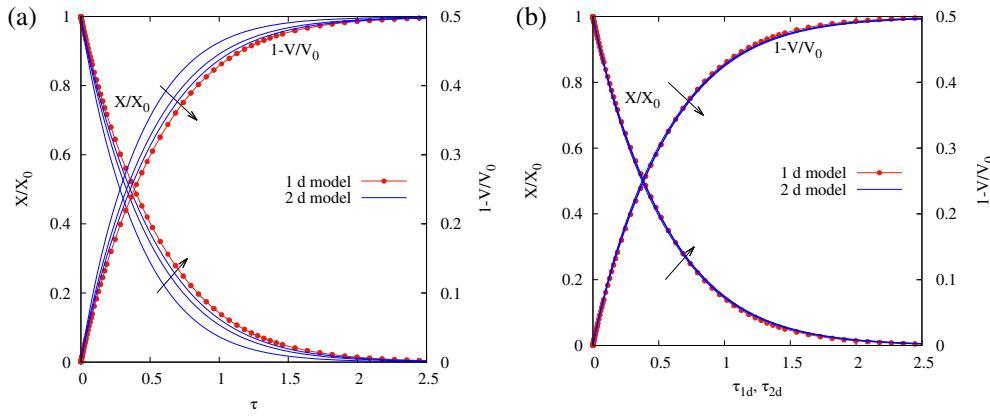


FIGURE 5 (a) X/X_0 and $1 - V/V_0$ versus $\tau = tD/R_0^2$ for a cylindrical sample with $L_0 > R_0$ for increasing values of the aspect ratio $L_0/R_0 = 2.5, 5, 10$. Arrows indicate increasing values of L_0/R_0 . Comparison between the 1-d model (insensitive to the aspect ratio, red dots) and the 2-d model (continuous lines) for $\alpha(\psi) = C = 1$, $\phi_0 = 0.5$, $Bi_m = 1$; (b) X/X_0 and $1 - V/V_0$ versus $\tau_{2d} = tD_{2d}/R_0^2$ for the 2-d model with $L_0/R_0 = 2.5, 5, 10$ and $D_{2d} = D_{1d}/\beta(L_0/R_0)$ collapsing onto the corresponding curves X/X_0 and $1 - V/V_0$ versus $\tau_{1d} = tD_{1d}/R_0^2$ for $\beta(2.5) = 1.31$, $\beta(5) = 1.17$, and $\beta(10) = 1.08$

model, we neglect (a) the shrinkage along the direction associated with the larger dimension and (b) the water flux from the surface orthogonal to this direction.

In this section we investigate and quantify the errors deriving from adopting a 1-d approach for modeling an intrinsic 2-d axial symmetric problem. Obviously a 1-d model can be adopted if and only if we are not interested in describing the actual surface deformation induced by the drying process. Such a deformation is more significant for larger values of Bi_m , as discussed in detail in Adrover et al. (2019b).

Our preliminary analysis focuses on the case of a constant and unitary shrinking factor $\alpha(\psi) = C = 1$, corresponding to the case in which the (dimensionless) shrinkage velocity is $\mathbf{v} = \phi_0 \nabla \psi$ and volume reduction equals the volume of removed water (ideal shrinkage).

Figure 1 shows the contour plots of rescaled water volume fraction $\psi(\rho, \zeta)$ for $X/X_0 = 0.1$ and for increasing values of $Bi_m = 1, 3, 5$ (from top to bottom). Both discoidal samples (left panel $R_0/L_0 = 5$) and cylindrical samples (right panel, $L_0/R_0 = 5$) are considered. Figures 1 show how, even for an high aspect ratio, a significant surface deformation occurs, that cannot be captured by a 1-d model.

From a quantitative point of view, Figure 2a,b shows the behavior of the thickness $\tilde{L} = L/L_0$ versus τ as obtained from the 1-d model Equations (11)–(14) (dotted lines) compared to the average thickness $\tilde{L}_{av} = L_{av}/L_0$ obtained from the two dimensional model Equations (5)–(7) for a discoidal sample with increasing aspect ratio $R_0/L_0 = 2.5, 5, 7.5, 10$ and for $Bi_m = 1$ (A) and $Bi_m = 5$ (B). Thickness averaging (along the radial direction) for the 2-d model is necessary because of surface shape deformation occurring during the drying process.

It can be observed that the asymptotic thickness predicted by the 1-d model is smaller than that obtained from the 2-d model.

The error $\epsilon_L^\infty = \frac{|(L^\infty/L_0)_{1d} - (L^\infty/L_0)_{2d}|}{(L^\infty/L_0)_{2d}} = 11\%$ for $R_0/L_0 = 2.5$, and reduces to $\epsilon_L^\infty \leq 5\%$ for $R_0/L_0 \geq 5$, for both cases $Bi_m = 1$ and $Bi_m = 5$. Analogously, Figures 3a,b shows the behavior of the radius $\tilde{R} = R/R_0$ versus τ as obtained from the 1-d model Equations (17)–(19) (dotted lines) compared to the average radius $\tilde{R}_{av} = R_{av}/R_0$ (averaged along the

axial direction) from the two dimensional model Equations (5)–(7) for a cylindrical sample with increasing aspect ratio $L_0/R_0 = 2.5, 5, 7.5, 10$ and for $Bi_m = 1$ (A) and $Bi_m = 5$ (B). Also in this case, the asymptotic radius predicted by the 1-d model is smaller than that obtained from the 2-d model.

The error $\epsilon_R^\infty = \frac{|(R^\infty/R_0)_{1d} - (R^\infty/R_0)_{2d}|}{(R^\infty/R_0)_{2d}} \simeq 11\%$ for $L_0/R_0 = 2.5$, and reduces to $\epsilon_R^\infty \leq 5\%$ for $L_0/R_0 \geq 5$, for both cases $Bi_m = 1$ and $Bi_m = 5$.

Therefore, an aspect ratio greater than 5 guarantees, in both cases analyzed regardless of Bi_m , an error less than 5% for the asymptotic thickness or radius.

However, the main limitation in adopting a 1-d model is the error in the prediction of the diffusion coefficient.

Figure 4a shows, for a discoidal sample, the reduction of the free moisture fraction X/X_0 and the increase of $1 - V/V_0$ with the dimensionless time τ as obtained, by using the same diffusivity, from the 1-d model Equations (11)–(14) (dotted lines) and from the 2-d model, for increasing values of the aspect ratio $R_0/L_0 = 2.5, 5, 10$. As expected, the 1-d model exhibits a slower decay of the free moisture fraction since it neglects the water flow from the lateral surface. The ratio $(S_t + S_b)/S_l$ between the total area of exchanging top and bottom surfaces ($S_t = S_b$) and the area of the insulated lateral surface S_l equals the aspect ratio R_0/L_0 .

The 1-d curve X/X_0 versus $\tau = tD_{1d}/L_0^2$ as well as the 1-d curve $1 - V/V_0$ versus τ can be almost superimposed to the corresponding 2-d curve for a given aspect ratio R_0/L_0 by adopting a diffusivity value $D_{1d} = D_{2d}\beta(R_0/L_0)$ where $\beta(R_0/L_0) \geq 1$ is a factor depending on the aspect ratio R_0/L_0 and decreasing with the increase of the aspect ratio. Equivalently, the 2-d curve X/X_0 versus τ , for given aspect ratio R_0/L_0 , can be superimposed to the 1-d curve by adopting a diffusivity value $D_{2d} = D_{1d}/\beta(R_0/L_0)$.

Figure 4b shows how the curves X/X_0 and $1 - V/V_0$ versus $\tau_{2d} = tD_{2d}/L_0^2$ for the 2-d model with $R_0/L_0 = 2.5, 5, 10$ and $Bi_m = 1$ collapse onto the corresponding curves X/X_0 and $1 - V/V_0$ versus $\tau_{1d} = tD_{1d}/L_0^2$ for $\beta(2.5) = 1.21$, $\beta(5) = 1.1$, and $\beta(10) = 1.05$

FIGURE 6 L_{av}/L_0 and V/V_0 versus τ for discoidal samples for increasing values of the aspect ratio $R_0/L_0 = 2.5(a)$, $5(b)$, $7.5(c)$, $10(d)$; $\alpha(\psi) = C = 1$, $\phi_0 = 0.5$, $Bi_m = 1.5$. Continuous thick lines show the behavior of the model curve Equation (30)

corresponding infinitesimal volume $\pi R_0^2 \Delta z$. Equation (26) must be numerically solved together with the boundary conditions Equations (12) and (13) and Equation (14) for the evolution of the boundary.

Correspondingly, for a cylindrical sample, Equation (17) is modified as

$$\frac{\partial \psi}{\partial \tau} = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial \psi}{\partial \rho} (1 - \alpha(\psi) \phi_0 \psi) \right) - \frac{2Bi_m}{L_0/R_0} (\psi - \psi_{eq}), \quad \rho \in (0, \tilde{R}(\tau)) \quad (27)$$

where the factor $2/(L_0/R_0)$ represents, in dimensionless terms, the ratio between the infinitesimal top and bottom surface $2 \times (2\pi R_0 \Delta r)$ and the corresponding infinitesimal volume $2\pi R_0 L_0 \Delta r$. Equation (27) must be numerically solved together with the boundary conditions Equation (18) and Equation (19) for boundary temporal evolution.

Let us focus on the 1-d modified model for discoidal sample. The modified 1-d model satisfies a modified integral equation

$$\frac{d(X/X_0)}{d\tau} = \frac{1}{\phi_0 \alpha_L(\psi_b)} \frac{d(L/L_0)}{d\tau} - \frac{2Bi_m}{(R_0/L_0)} (X - X_{eq}) \quad (28)$$

Therefore, from experimental data L/L_0 versus X/X_0 , by adopting the modified model, it is no longer possible to estimate the shrinking factor $\alpha(\psi)$ from Equations (15) to (16), that is, independently of the diffusivity and transport parameters.

Nevertheless, in order to significantly reduce the error in the estimate of the diffusivity, the modified 1-d model can be still successfully applied by adopting the following numerical strategy:

1. From experimental calibration curve L/L_0 versus X/X_0 one preliminarily identifies the least-square best fit function $G_L(X/X_0)$, Equation (16).
2. By assuming a linear model $X/X_0 = g(\psi_b, Bi_m) \simeq \psi_b$ and therefore adopting $\alpha(\psi) = G_L'(\psi)/\phi_0$ in the 1-d model, thus neglecting the radial shrinkage and the water flux from the lateral surface Equations (11)–(14), one can obtain a more accurate estimate of the g function by a least-square best fit of numerical data X/X_0 versus ψ_b by means of the versatile sigmoidal function:

$$X/X_0 = g(\psi_b, Bi_m) = 1 - \frac{\kappa_1 (1 - \psi_b)^2}{1 + [\kappa_1 - 1 - \kappa_2] (1 - \psi_b)^n + \kappa_2 (1 - \psi_b)^2} \quad (29)$$

where κ_1 , κ_2 , and n are the best fit values for the corresponding Bi_m value.

3. With the more accurate expression for the shrinkage factor $\alpha(\psi) = G_L'(g[\psi, Bi_m])/ \phi_0$, one numerically solves the 1-d model Equations (11)–(14) thus verifying the accuracy of model predictions L/L_0 versus X/X_0 onto the experimental calibration curve $G_L(X/X_0)$

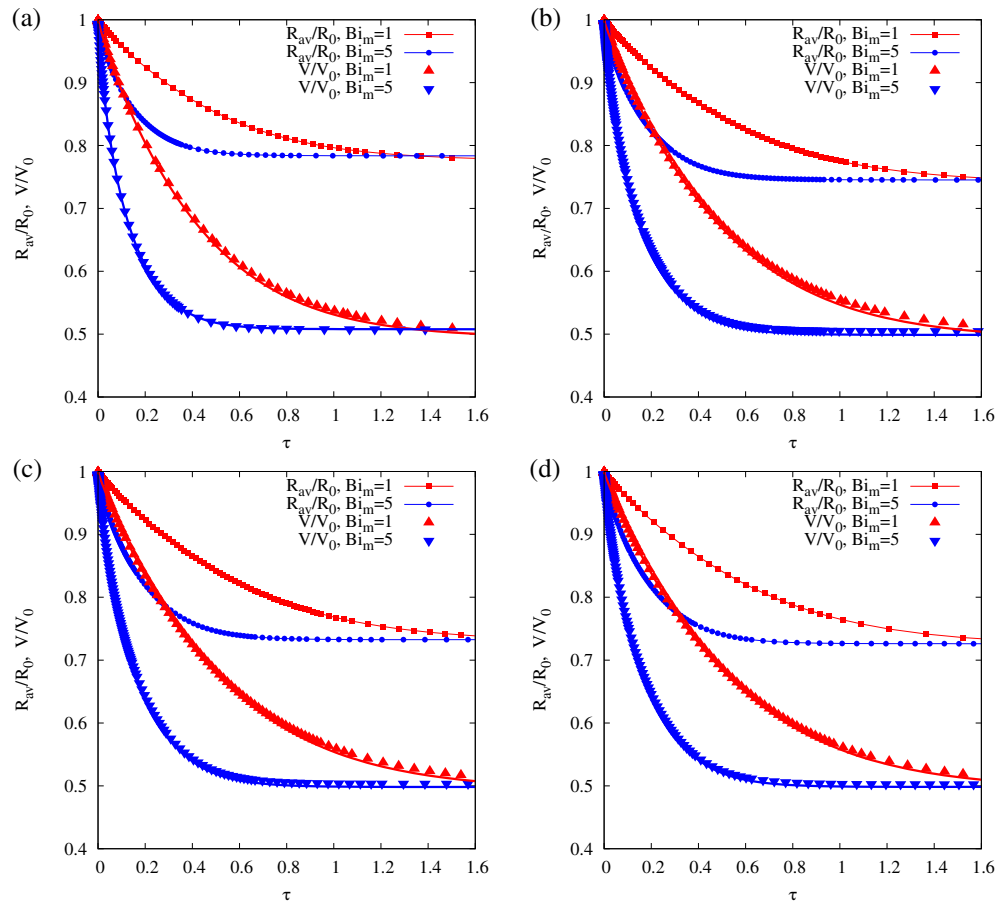


FIGURE 7 R_{av}/R_0 and V/V_0 versus τ for cylindrical samples for increasing values of the aspect ratio $L_0/R_0 = 2.5(a)$, $5(b)$, $7.5(c)$, $10(d)$; $\alpha(\psi) = C = 1$, $\phi_0 = 0.5$, $Bi_m = 1.5$. Continuous thick lines show the behavior of the model curve Equation (31)

4. With the above verified expression for the shrinkage factor $\alpha(\psi)$ one can finally numerically solve the 1-d modified model Equations (12)–(14) and Equation (26), thus obtaining a model curve X/X_0 versus τ accounting also for water flux from lateral surface and proceeds to the estimate of the diffusivity by means of a

least-square best fit of the model curve X/X_0 versus $\tau = tD/L_0^2$ onto experimental data X/X_0 versus t .

Exactly the same procedure can be adopted for the cylindrical sample case when the calibration curve R/R_0 versus X/X_0 is available together with experimental data X/X_0 versus t (see Equations (18)–(19) and Equation (27)).

6 | ESTIMATE OF SAMPLE VOLUME FROM SAMPLE THICKNESS OR RADIUS

In order to verify and quantify the accuracy of 1-d model predictions one can adopt a 2-d model for the comparison. If only experimental data on sample thickness L/L_0 or sample radius R/R_0 are available, in order to have a fair comparison between 1-d and 2-d models, one preliminary needs to estimate the actual volume of the sample associated with the experimental thickness or radius, thus extrapolating the calibration curve V/V_0 versus X/X_0 necessary for estimating the shrinkage factor $\alpha_V(X/X_0)$ for the 2-d model.

Figure 6a–d shows the behavior of the average rescaled thickness L_{av}/L_0 and the rescaled volume V/V_0 versus τ as obtained from a 2-d model for a discoidal sample in the test case $\alpha = C = 1$, $\phi_0 = 0.5$ for

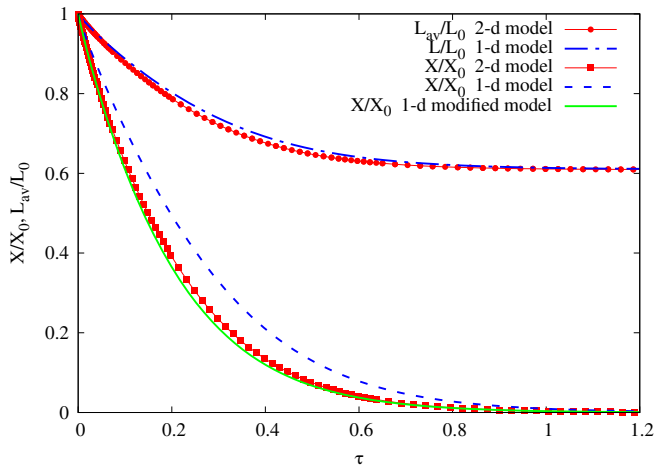


FIGURE 8 L_{av}/L_0 and X/X_0 versus τ for the three models analyzed with $\phi_0 = 0.51$, $Bi_m = 2$. For 2-d model, $\alpha_V(\psi) = C_V = 0.858$; for 1-d and 1-d modified models, $\alpha_L(\psi) = C_L = 0.765$

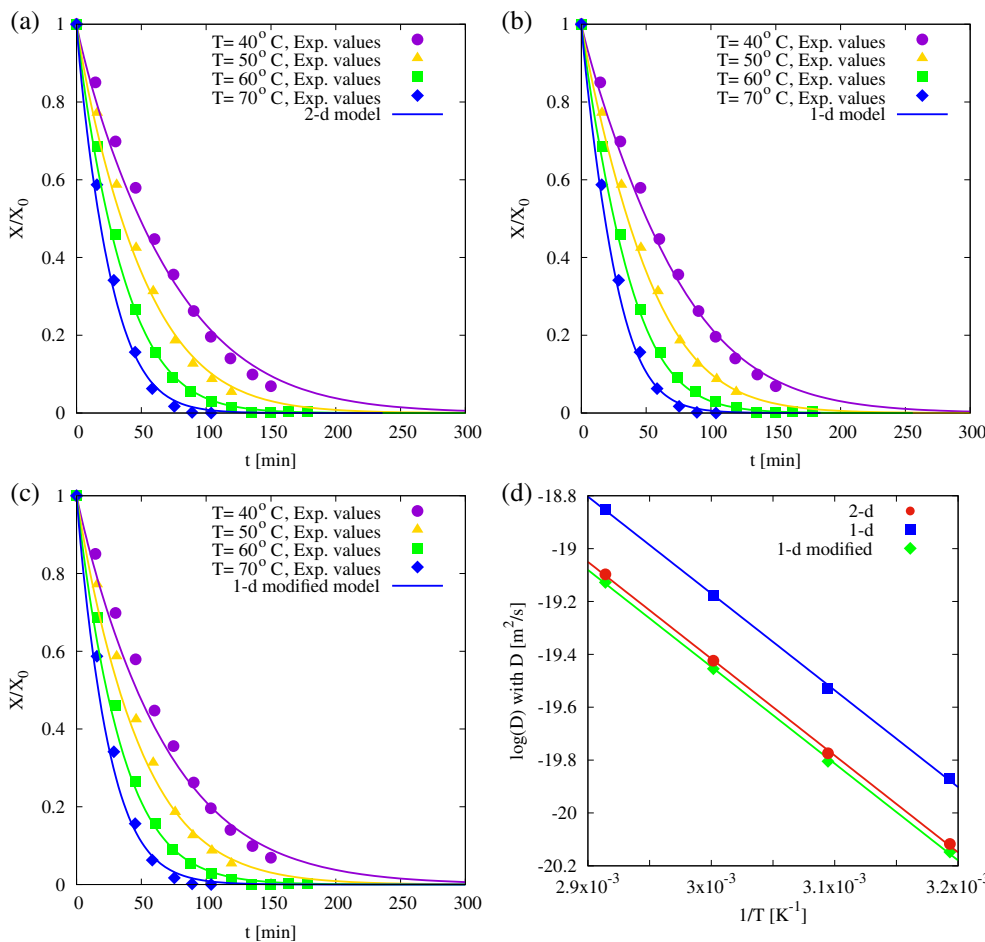


FIGURE 9 X/X_0 versus time t (min) for eggplant discoidal samples with initial dimensions $R_0 = 15$ mm, $L_0 = 6$ mm. (a) 2-d model, $\alpha_V(\psi) = C_V = 0.858$, $\phi_0 = 0.51$, $Bi_m = 2$; (b) 1-d model, $\alpha_L(\psi) = C_L = 0.765$, $\phi_0 = 0.51$, $Bi_m = 2$; (c) modified 1-d model, $\alpha_L(\psi) = C_L = 0.765$, $\phi_0 = 0.51$, $Bi_m = 2$. (d) Diffusivities estimated from a least-square best fit of experimental data with the 2-d model, the 1-d model and the 1-d modified model. Straight lines represent the Arrhenius correlation function $D(T) = D_0 \exp[-E/(RT)]$ with $E/R = 3,662.98$ K and $D_0 = 2.187 \times 10^{-4}$ m²/s for the 2-d model, $D_0 = 2.799 \times 10^{-4}$ m²/s for the 1-d model, $D_0 = 2.143 \times 10^{-4}$ m²/s for the 1-d modified model

increasing values of the aspect ratio $R_0/L_0 = 2.5, 5, 7.5, 10$ and for two values of $Bi_m = 1, 5$. Continuous lines represent the expression

$$\frac{V}{V_0} = \frac{L_{av}}{L_0} \left[1 - \frac{1}{4} \left(\frac{L_0}{R_0} \right) \left(1 - \frac{L_{av}}{L_0} \right) \right]^2, \quad (30)$$

relating the actual sample volumes to the average discoidal sample thicknesses (averaged along the radial direction). It is worth noting the good agreement between these curves and 2-d model data for V/V_0 versus τ .

The expression Equation (30) has been obtained by observing, from 2-d model data, that $\Delta R = R_0 - R_{av} \simeq \Delta L/4 = (L_0 - L_{av})/4$ at each time instant. The agreement is quite satisfactory even at low values of the aspect ratio, where the difference between V/V_0 and L/L_0 is more pronounced.

Similarly, Figure 7a–d show the behavior of the average rescaled radius R_{av}/R_0 and the rescaled volume V/V_0 versus τ as obtained from a 2-d model for a long-thin cylinder in the test case $\alpha = C = 1$, $\phi_0 = 0.5$ for increasing values of the aspect ratio $L_0/R_0 = 2.5, 5, 7.5, 10$ and for two values of $Bi_m = 1, 5$. Continuous lines refer to the following expression:

$$\frac{V}{V_0} = \left(\frac{R}{R_0} \right)^2 \left[1 - 2 \left(\frac{R_0}{L_0} \right) \left(1 - \frac{R_{av}}{R_0} \right) \right] \quad (31)$$

relating the actual sample volumes to the average sample radii (averaged along the radial direction). The good agreement between 2-d model data for V/V_0 versus τ and the expression Equation (31) is observed.

The expression Equation (31) has been obtained by observing, from 2-d model data, that $\Delta R = R_0 - R_{av} \simeq \Delta L/2 = (L_0 - L_{av})/2$ at each time instant for a long-thin cylinder.

In the next two Sections 7 and 8 we analyze experimental data obtained from convective drying of discoidal samples of eggplants and chayote by making use of both 1-d and 2-d approaches and quantifying 1-d modeling errors in two typical real cases characterized by $\alpha(\psi) \neq 1$. We also apply the modified 1-d model to quantify the improvement in diffusivity estimation.

7 | EGGPLANT DISCOIDAL SAMPLES

We focus on experimental data of convective hot-air drying of eggplant discoidal samples (*Solanum melongena*, Longo cultivar, data from Brasiello et al. (2013)). Vegetables were washed and peeled. Cylindrical slices with initial radius $R_0 = 15$ mm and initial thickness $L_0 = 6$ mm were prepared. The discoidal sample aspect ratio is $R_0/L_0 = 2.5$. Eggplant drying (at $T = 40, 50, 60, 70^\circ\text{C}$) was carried out in a convective dryer. The initial moisture content was 12.704 ± 0.04 kg/kg dry basis. The initial water volume fraction was $\phi_0 = 0.51$ (data from Food Composition Databases of USDA US Department of Agriculture, <https://ndb.nal.usda.gov/ndb/>).

Available experimental data are the free moisture fraction X/X_0 versus time (min) at the four temperatures analyzed and the average sample thickness at the end of the drying process. The asymptotic average thickness seems to be almost independent of the drying temperature and can be approximated as $L_{av}^\infty \simeq 0.61 \pm 0.03$ mm.

Since we have no information regarding the temporal evolution of the sample thickness, and therefore no calibration curve L/L_0 versus X/X_0 is available, we assume for the 1-d model $\alpha_L(\psi) = C_L$ where the constant C_L is set from the asymptotic thickness by using a linear law:

$$\frac{L_{av}^\infty}{L_0} \simeq 0.61 = 1 - C_L \phi_0 \rightarrow C_L = \frac{1}{\phi_0} \left(1 - \frac{L_{av}^\infty}{L_0} \right) \simeq 0.765 \quad (32)$$

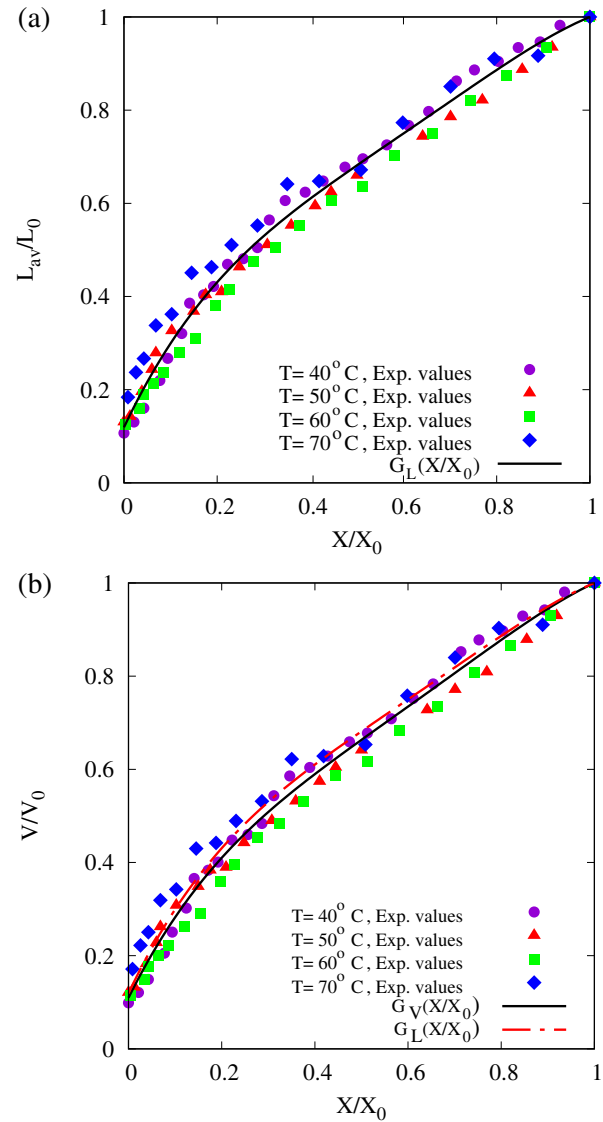


FIGURE 10 (a) Experimental data L_{av}/L_0 versus X/X_0 for chayote's discoidal samples at four different temperatures $T = 40, 50, 60, 70^\circ\text{C}$. (b) V/V_0 versus X/X_0 as obtained from experimental data L_{av}/L_0 versus X/X_0 and Equation (30). Continuous and dashed lines represent the least-square fit fourth order polynomials $G_L(X/X_0)$, Equation (35) and $G_V(X/X_0)$, Equation (36)

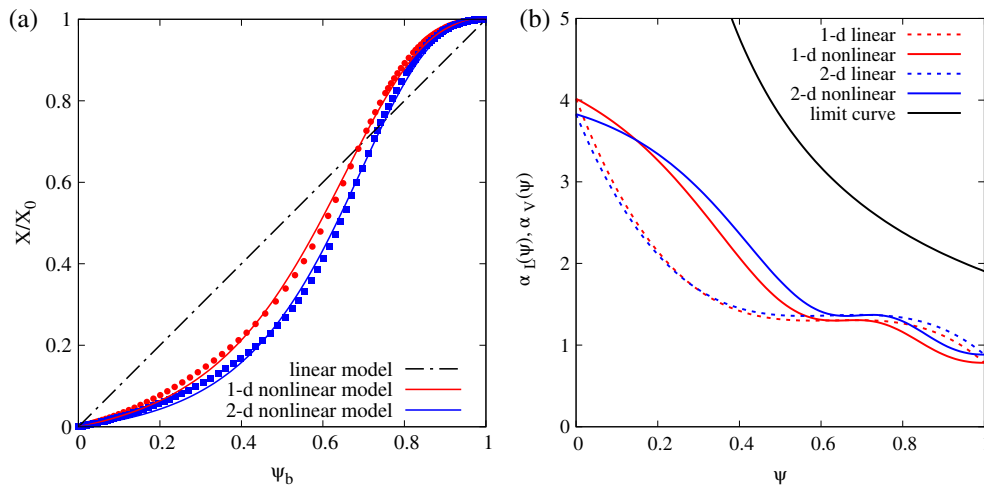


FIGURE 11 (a) X/X_0 versus ψ_b obtained from the numerical integration of the 1-d and the 2-d models with $\alpha_L(\psi) = G_L'(\psi)/\phi_0$ and $\alpha_V(\psi) = G_V'(\psi)/\phi_0$, respectively. For both models $Bi_m = 2$. Continuous lines represent the best fit curves $X/X_0 = g_L(\psi_b, Bi_m)$ and $X/X_0 = g_V(\psi_b, Bi_m)$ (see Equation (29)) with $\kappa_1 = 0.247$, $\kappa_2 = 0.243$, and $n = 0.044$ for the 1-d model and $\kappa_1 = 0.472$, $\kappa_2 = 0.477$, and $n = 0.069$ for the 2-d model. (b) Thickness shrinkage factor $\alpha_L(\psi)$ and volume shrinkage factor $\alpha_V(\psi)$ with the linear approximation $X/X_0 \simeq \psi_b$, that is, $\alpha_L(\psi) = G_L'(\psi)/\phi_0$ and $\alpha_V(\psi) = G_V'(\psi)/\phi_0$ and the nonlinear approximation $X/X_0 = g_L(\psi_b, Bi_m)$ and $X/X_0 = g_V(\psi_b, Bi_m)$, that is, $\alpha_L(\psi) = G_L'(g_L[\psi, Bi_m])/\phi_0$ and $\alpha_V(\psi) = G_V'(g_V[\psi, Bi_m])/\phi_0$. Continuous black line represents the limit curve $\alpha(\psi) = (\phi_0\psi)^{-1}$

For the 2-d model we set $\alpha_V = C_V$ where the constant C_V is set from the asymptotic thickness according to Equation (30), relating average rescaled thickness L_{av}/L_0 to the rescaled volume V/V_0

$$\frac{V^\infty}{V_0} = \frac{L_{av}^\infty}{L_0} \left[1 - \frac{1}{4} \left(\frac{L_0}{R_0} \right) \left(1 - \frac{L_{av}^\infty}{L_0} \right) \right]^2 \simeq 0.563 \quad (33)$$

$$\frac{V^\infty}{V_0} = 1 - C_V \phi_0 \rightarrow C_V \simeq 0.856 \quad (34)$$

This choice for α_L and α_V guarantees that both the 1-d and the 2-d models exhibit the same asymptotic average thickness L_{av}^∞/L_0 .

Since we assumed, in the absence of experimental data, $\alpha_L(\psi)$ and $\alpha_V(\psi)$ independent of ψ_b , there is no need to apply steps 2 and 3 of the numerical procedure described in Section 4 and we can directly proceed to numerically integrate the 1-d model with $\alpha_L = C_L$, the 1-d modified model with the same $\alpha_L = C_L$ and the 2-d model with $\alpha_V = C_V$.

Figure 8 shows the behavior of L_{av}/L_0 versus τ as obtained from the 1-d and the 2-d models. The same figure also shows the behavior of X/X_0 versus τ as obtained from the 1-d, the 1-d modified and the 2-d models.

The curve X/X_0 versus τ obtained from the 1-d model is the slower one, while those obtained from the 1-d modified and the 2-d models are very close to each other, leading to very close values of the estimated diffusivities.

Figure 9a–c shows the comparison between experimental data for X/X_0 versus t and data predicted from the three different models analyzed. The accuracy of the models in describing experimental data is fully satisfactory, but the 1-d model predicts diffusivities that are larger than diffusivities predicted by the 2-d model by factor 1.28, thus implying an overestimation error $\varepsilon_D = 28\%$. On the contrary, the percentage error for the diffusivity predicted by the 1-d modified model is extremely small and equal to $\varepsilon_D = 2\%$ (see Figure 9d). For the

2-d model estimated diffusivities are $D(40^\circ\text{C}) = 1.83 \times 10^{-9} \text{ m}^2/\text{s}$, $D(50^\circ\text{C}) = 2.58 \times 10^{-9} \text{ m}^2/\text{s}$, $D(60^\circ\text{C}) = 3.66 \times 10^{-9} \text{ m}^2/\text{s}$, $D(70^\circ\text{C}) = 5.08 \times 10^{-9} \text{ m}^2/\text{s}$ with an Arrhenius correlation function $D(T) = D_0 \exp[-E/(RT)]$ with $D_0 = 2.187 \times 10^{-4} \text{ m}^2/\text{s}$ and $E/R = 3,662.98 \text{ K}$. For diffusivities estimated with the 1-d model and the 1-d modified model, the activation energy is the same as for the 2-d model $E/R = 3,662.98 \text{ K}$ and $D_0 = 2.799 \times 10^{-4} \text{ m}^2/\text{s}$ for 1-d model and $D_0 = 2.143 \times 10^{-4} \text{ m}^2/\text{s}$ for the 1-d modified model

8 | CHAYOTE DISCOIDAL SAMPLES

We further focus on experimental data of convective hot-air drying of chayote discoidal samples (data from Ruiz-López et al. (2012)). Fruits were washed and peeled. Cylindrical slices with initial radius $R_0 = 35 \text{ mm}$ and initial thickness $L_0 = 6 \text{ mm}$ were prepared. The discoidal aspect ratio is $R_0/L_0 \simeq 5.83$. Chayote drying (at $T = 40, 50, 60, 70^\circ\text{C}$) was carried out in a convective dryer with air velocity 2 m/s . The initial moisture content was $93.38 \pm 1.03 \text{ g water}/100 \text{ g product}$. The initial water volume fraction was $\phi_0 = 0.525$. Available experimental data are the free moisture fraction X/X_0 versus time (min) and the dehydration “calibration” curve L_{av}/L_0 versus X/X_0 at the four temperatures analyzed.

Figure 10a shows experimental data for L_{av}/L_0 versus X/X_0 and the least-square best fit of the fourth-order polynomial function G_L

$$G_L(X/X_0) = 1 - \sum_{j=1}^4 \beta_j^L (1 - X/X_0)^j \quad (35)$$

$$\beta_1^L = 0.411, \beta_2^L = 1.155, \beta_3^L = -2.131, \beta_4^L = 1.445$$

It can be observed that the dehydration curve is almost independent of the drying temperature T and therefore a unique $G_L(X/X_0)$

function (and a unique $\alpha_L(\psi)$ thickness shrinkage function) describes the entire set of experimental data at different temperatures.

Figure 10b shows the calibration curve V/V_0 versus X/X_0 and the corresponding least-square best fit fourth-order polynomial function G_V

$$G_V(X/X_0) = 1 - \sum_{j=1}^4 \beta_j^V (1 - X/X_0)^j \quad (36)$$

$$\beta_1^V = 0.464, \beta_2^V = 1.092, \beta_3^V = -2.025, \beta_4^V = 1.359$$

estimated from L_{av}/L_0 versus X/X_0 data and Equation (30) relating V/V_0 to L_{av}/L_0 in a discoidal sample. Figure 10b shows both the G_V and the G_L functions in order to highlight how, in the case of an high aspect ratio $R_0/L_0 > 5$, the rescaled volume and the rescaled thickness are quantitatively very close to each other. Anyhow, we still account for these small differences and proceed on parallel tracks in the application of the 1-d and the 2-d models.

The calibration curves shown in Figure 10a,b are highly nonlinear, and therefore we need to adopt the procedure described in Section 4 in order to estimate the correct thickness shrinkage factor $\alpha_L(\psi)$ for the 1-d and the 1-d modified model, and the volume shrinkage factor $\alpha_V(\psi)$ for the 2-d model. Given the high value of the discoidal aspect ratio $R_0/L_0 \simeq 5.83$ we expect a relatively small percentage error, around 10% for the diffusivity estimated with the 1-d model with respect to that obtained with the 2-d model. Our goal is to verify this expectation also in the case of a highly nonlinear dehydration curve.

Step 2 of the numerical procedure requires the numerical integration of the moving boundary models with $\alpha_L(\psi) = G_L'(\psi)/\phi_0$ for the 1-d model and $\alpha_V(\psi) = G_V'(\psi)/\phi_0$ (this implying the adoption of the linear model $X/X_0 \simeq \psi_b$) in order to obtain a numerical estimation of the curve $X/X_0 = g(\psi_b, Bi_m)$ where ψ_b is the rescaled water volume fraction at the probe point initially placed at $\rho = 1$ in the 1-d model and $(\rho, \zeta) = (1, 0)$ for the 2-d model.

Figure 11 shows the numerical results for X/X_0 versus ψ_b obtained from the numerical integration of the 1-d and the 2-d models with $\alpha_L(\psi) = G_L'(\psi)/\phi_0$ and $\alpha_V(\psi) = G_V'(\psi)/\phi_0$, respectively. For both models $Bi_m = 2$. Continuous lines represent the best fit curves $X/X_0 = g_L(\psi_b, Bi_m)$ and $X/X_0 = g_V(\psi_b, Bi_m)$ (see Equation (29)) with $\kappa_1 = 0.247$, $\kappa_2 = 0.243$, and $n = 0.044$ for the 1-d model and $\kappa_1 = 0.472$, $\kappa_2 = 0.477$, and $n = 0.069$ for the 2-d model. Correspondingly Figure 11b shows the thickness shrinkage factor $\alpha_L(\psi)$ and the volume shrinkage factor $\alpha_V(\psi)$ with the linear approximation $X/X_0 \simeq \psi_b$, that is, $\alpha_L(\psi) = G_L'(\psi)/\phi_0$ and $\alpha_V(\psi) = G_V'(\psi)/\phi_0$ and the nonlinear approximation $X/X_0 = g_L(\psi_b, Bi_m)$ and $X/X_0 = g_V(\psi_b, Bi_m)$, that is, $\alpha_L(\psi) = G_L'(g_L[\psi, Bi_m])/\phi_0$ and $\alpha_V(\psi) = G_V'(g_V[\psi, Bi_m])/\phi_0$.

Figure 11b also shows the limit curve $\alpha(\psi) = (\phi_0\psi)^{-1}$ derived in Adrover et al. (2019a) in order to highlight how both shrinkage functions $\alpha_L(\psi)$ and $\alpha_V(\psi)$ satisfy the physical constrain $\alpha(\psi) \leq (\phi_0\psi)^{-1}$, $\forall \psi \in [0, 1]$. For a detailed discussion of this issue see Adrover et al. (2019a)

In the Step 3, we numerically integrate both the 1-d and the 2-d models with the more accurate nonlinear expression for $\alpha_L(\psi)$ and

$\alpha_V(\psi)$ evaluated at Step 2 and verify the accuracy of model predictions for L_{av}/L_0 versus X/X_0 onto the corresponding experimental curve. Figure 12a shows the comparison between the experimental curve L_{av}/L_0 versus X/X_0 and 1-d and 2-d model predictions. Red dashed line represents 1-d model prediction with the linear assumption $X/X_0 \simeq \psi_b$ and therefore with the less accurate expression for the thickness shrinkage factor $\alpha_L(\psi) = G_L'(\psi)/\phi_0$. Continuous blue line and black dotted line (almost coinciding) represent 1-d and 2-d model predictions with the more accurate nonlinear functions $\alpha_L(\psi) = G_L'(g_L[\psi, Bi_m])/\phi_0$ and $\alpha_V(\psi) = G_V'(g_V[\psi, Bi_m])/\phi_0$.

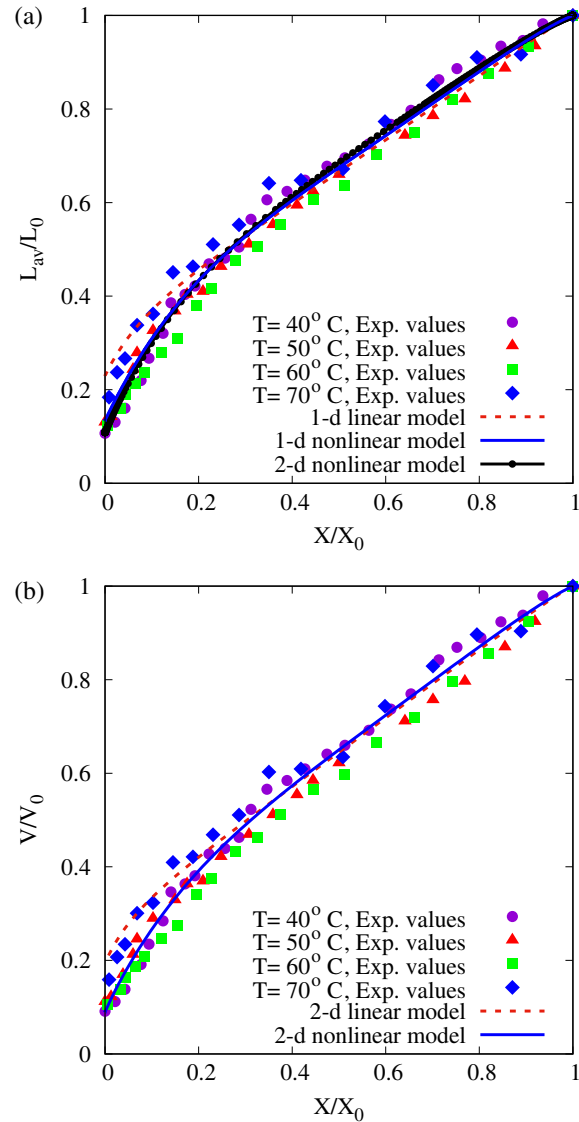


FIGURE 12 (a) Experimental data L_{av}/L_0 versus X/X_0 for chayote discoidal samples at four different temperatures $T = 40, 50, 60, 70^\circ\text{C}$. (b) V/V_0 versus X/X_0 as obtained from experimental data L_{av}/L_0 versus X/X_0 and Equation (30). Red dashed lines represent 1-d and 2-d model predictions with the linear model $X/X_0 \simeq \psi_b$, that is, $\alpha_L(\psi) = G_L'(\psi)/\phi_0$ and $\alpha_V(\psi) = G_V'(\psi)/\phi_0$, respectively. Blue continuous lines represent 1-d and 2-d model predictions with the nonlinear model $\alpha_L(\psi) = G_L'(g_L[\psi, Bi_m])/\phi_0$ and $\alpha_V(\psi) = G_V'(g_V[\psi, Bi_m])/\phi_0$. Black dotted line represents rescaled thickness predictions of the 2-d model with the nonlinear function $\alpha_V(\psi) = G_V'(g_V[\psi, Bi_m])/\phi_0$

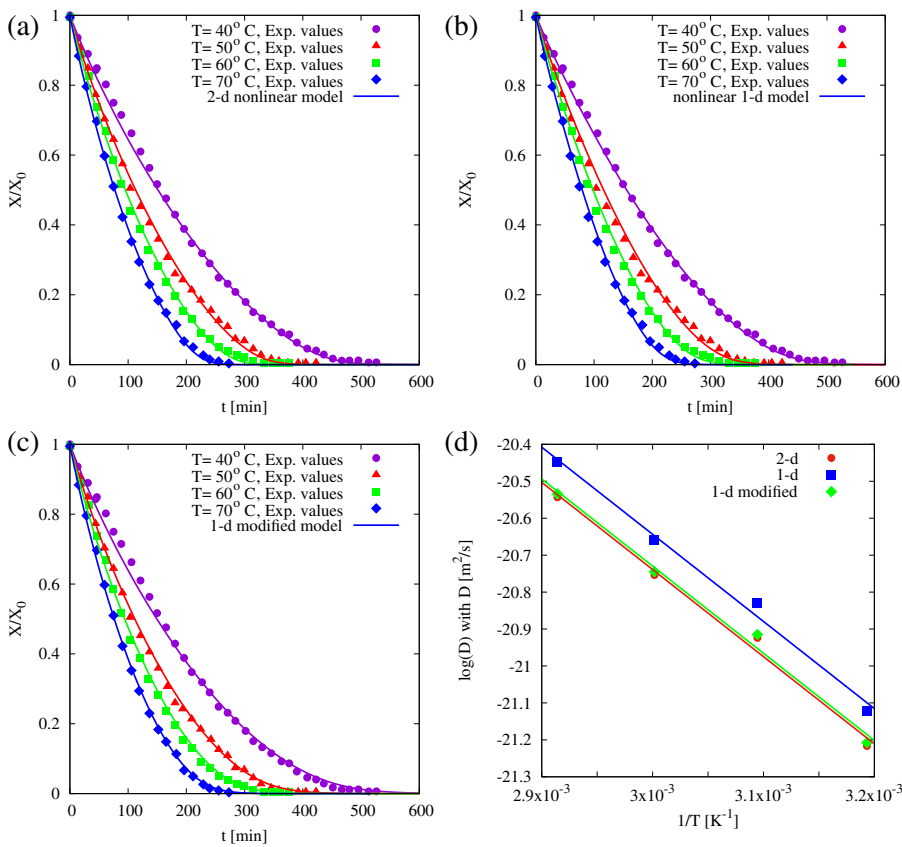


FIGURE 13 X/X_0 versus time t (min) for chayote's discoidal samples with initial dimensions $R_0 = 35$ mm, $L_0 = 6$ mm. (a) 2-d model; (b) 1-d model; (c) modified 1-d model (d) Diffusivities estimated from a least-square best fit of experimental data with the 2-d model, the 1-d model and the 1-d modified model. Straight lines represent the Arrhenius correlation function $D(T) = D_0 \exp[-E/(RT)]$ with $E/R = 2,358.38$ K and $D_0 = 1.164 \times 10^{-6}$ m^2/s for the 2-d model, $D_0 = 1.28 \times 10^{-6}$ m^2/s for the 1-d model, $D_0 = 1.174 \times 10^{-6}$ m^2/s for the 1-d modified model

Figure 12b shows comparison between the experimental curve V/V_0 versus X/X_0 and the 2-d model predictions with the less accurate and the more accurate nonlinear model for the volume shrinkage function $\alpha_V(\psi)$.

For both the 1-d and the 2-d models, the improvement obtained by adopting the more accurate expression for the shrinkage factor is particularly important for lower values of the free moisture fraction $X/X_0 \leq 0.3$.

In Step 4 we can now numerically integrate also the 1-d modified model with the more accurate expression for $\alpha_L(\psi)$ evaluated in the Step 2 and verified in the Step 3. We therefore have three different curves X/X_0 versus τ obtained from the 1-d, the 1-d modified and the 2-d models and we can proceed to estimate the diffusivities with each model by a least-square best fit of experimental data X/X_0 versus t . Figure 13a–c shows the comparison between experimental data for X/X_0 versus t and the three different models analyzed. The accuracy of all the three models in describing experimental data is fully satisfactory, but the 1-d model predicts diffusivities that are larger than diffusivities predicted by the 2-d model by factor 1.1, thus implying an overestimation error $\varepsilon_D = 11\%$. On the contrary, the percentage error for the diffusivity predicted by the 1-d modified model is extremely small and equal to $\varepsilon_D = 1\%$ (see Figure 9d). For the 2-d model estimated diffusivities are $D(40^\circ\text{C}) = 6.1 \times 10^{-10}$ m^2/s , $D(50^\circ\text{C}) = 8.18 \times 10^{-10}$ m^2/s , $D(60^\circ\text{C}) = 9.69 \times 10^{-10}$ m^2/s , $D(70^\circ\text{C}) = 1.19 \times 10^{-9}$ m^2/s with an Arrhenius correlation function $D(T) = D_0 \exp[-E/(RT)]$ with $D_0 = 1.164 \times 10^{-6}$ m^2/s and $E/R = 2,358.38$ K. For diffusivities estimated with the 1-d model and the 1-d modified

model, the activation energy is the same as for the 2-d model $E/R = 2,358.38$ K and $D_0 = 1.28 \times 10^{-6}$ m^2/s for the 1-d model, $D_0 = 1.174 \times 10^{-6}$ m^2/s for the 1-d modified model. Diffusivity estimated from the 2-d model are comparable with literature data for other vegetable products as carrots (2.58×10^{-10} to 1.72×10^{-9} m^2/s , 60–90°C), mango (2.61×10^{-10} to 1.30×10^{-9} m^2/s , 40–70°C), potatoes (3.55×10^{-10} to 1.92×10^{-9} m^2/s , 40–85°C) in which shrinkage has also been considered (Dissa, Desmorieux, Bathiebo, & Koulidiati, 2008; Hassini, Azzouz, Pecalski, & Belghith, 2007; Zielinska & Markowski, 2010).

9 | CONCLUSIONS

In this article, the comparison between 1-d and 2-d versions of the mathematical model developed in Adrover, et al. (2019a, 2019b) is discussed. The model describes mass transport in foods when volume changes occur.

The novelty of the model lies in a constitutive equation linking shrinkage velocity and water concentration gradient through a proportionality function α which can be determined through experiments. Being α related to the rescaled volume of the sample and dependent on water concentration, the knowledge of the proportionality function α is thus strictly connected to the possibility of measuring volume changes.

In many practical cases, irregular shapes prevent measurements of the whole volumes. When this happens, being possible to deal only

with the dimensional variation along one main dimension, the calculation of α is accomplished accordingly. Hence, the applicability of the model could be compromised since a no satisfactory estimate of α implies errors in transport parameters estimate.

In order to assess this issue, we addressed the cases of discoidal and long-thin cylindrical samples by using both the 1-d and 2-d approaches, thus investigating the influence of Biot number and aspect ratio. We showed that the use of the 1-d approach causes an overestimation of the transport parameter (namely the water diffusion coefficient) as a consequence of neglecting significant portions of the exchange surface. The magnitude of the overestimation depends on the aspect ratio of the sample only, regardless the Biot number; even in the case of high aspect ratios (about 5), the error in the diffusion coefficient estimate is about 11% for discoidal samples and 17% for long-thin cylindrical samples.

In order to avoid this serious issue, we proposed a method based on the introduction of an additional term into the model's equation of drying which compensates the neglected water flux, thus reducing the error in diffusion coefficient estimate below 2%. The developed method is applied to two cases taken from literature (i.e., eggplant and chayote drying).

The numerical strategy proposed for the implementation of the 1-d modified model has been successfully applied to chayote's air-drying experimental data, characterized by an high nonlinear calibration curve. This observation supports the idea that the method can be in principle applied to any food material, significantly improving prediction capabilities of the 1-d model.

CONFLICT OF INTEREST

Authors have no conflict of interest to declare.

ORCID

Alessandra Adrover  <https://orcid.org/0000-0001-5939-740X>

Antonio Brasiello  <https://orcid.org/0000-0002-3622-5258>

REFERENCES

- Adrover, A., & Brasiello, A. (2019). A moving boundary model for isothermal drying and shrinkage of chayote discoid samples: Comparison between the fully analytical and the shortcut numerical approaches. *International Journal of Chemical Engineering*, 2019, 3926897.
- Adrover, A., Brasiello, A., & Ponso, G. (2019a). A moving boundary model for food isothermal drying and shrinkage: A shortcut numerical method for estimating the shrinkage factor. *Journal of Food Engineering*, 244, 212–219.
- Adrover, A., Brasiello, A., & Ponso, G. (2019b). A moving boundary model for food isothermal drying and shrinkage: General setting. *Journal of Food Engineering*, 244, 178–191.
- Ansari, I., & Datta, A. (2003). An overview of sterilization methods for packaging materials used in aseptic packaging systems. *Food and Bioprocess Processing*, 81(1), 57–65.
- Aprajeeta, J., Gopirajah, R., & Anandharamakrishnan, C. (2015). Shrinkage and porosity effects on heat and mass transfer during potato drying. *Journal of Food Engineering*, 144, 119–128.
- Bird, R. B., Stewart, W. E., & Lightfoot, E. N. (2002). *Transport phenomena* (2nd ed.). New York, NY: John Wiley & Sons, Inc.
- Brasiello, A., Adiletta, G., Russo, P., Crescitelli, S., Albanese, D., & Di Matteo, M. (2013). Mathematical modeling of eggplant drying: Shrinkage effect. *Journal of Food Engineering*, 114(1), 99–105.
- Brasiello, A., Crescitelli, S., & Giona, M. (2016). One-dimensional hyperbolic transport: Positivity and admissible boundary conditions derived from the wave formulation. *Physica A: Statistical Mechanics and its Applications*, 449, 176–191.
- Brasiello, A., Crescitelli, S., & Milano, G. (2012). A multiscale approach to triglycerides simulations: From atomistic to coarse-grained models and back. *Faraday Discussions*, 158, 479–492.
- Brasiello, A., Iannone, G., Adiletta, G., De Pasquale, S., Russo, P., & Di Matteo, M. (2017). Mathematical model for dehydration and shrinkage: Prediction of eggplant's mri spatial profiles. *Journal of Food Engineering*, 203, 1–5.
- Brasiello, A., Russo, L., Siettos, C., Milano, G., & Crescitelli, S. (2010). Multi-scale modelling and coarse-grained analysis of triglycerides dynamics. *Computer Aided Chemical Engineering*, 28(C), 625–630.
- Crank, J. (1975). *The mathematics of diffusion*. Oxford, UK: Oxford University Press.
- Curcio, S., & Aversa, M. (2014). Influence of shrinkage on convective drying of fresh vegetables: A theoretical model. *Journal of Food Engineering*, 123, 36–49.
- Datta, A. (2007). Porous media approaches to studying simultaneous heat and mass transfer in food processes. I: Problem formulations. *Journal of Food Engineering*, 80(1), 80–95.
- Defraeye, T. (2014). Advanced computational modelling for drying processes—A review. *Applied Energy*, 131, 323–344.
- Dissa, A., Desmorieux, H., Bathiebo, J., & Koulidiati, J. (2008). Convective drying characteristics of amélie mango (*Mangifera indica* L. cv. 'Amélie') with correction for shrinkage. *Journal of Food Engineering*, 88(4), 429–437.
- Frenkel, D., & Smit, B. (1996). *Understanding molecular simulation. From algorithm to applications*. San Diego, CA: Academic Press.
- Giona, M., Brasiello, A., & Crescitelli, S. (2015). Ergodicity-breaking bifurcations and tunneling in hyperbolic transport models. *EPL*, 112(3), 30001.
- Giona, M., Brasiello, A., & Crescitelli, S. (2017a). Stochastic foundations of undulatory transport phenomena: Generalized Poisson-Kac processes – Part ii irreversibility, norms and entropies. *Journal of Physics A: Mathematical and Theoretical*, 50(33), 335003. <https://doi.org/10.1088/1751-8121/aa79c5>
- Giona, M., Brasiello, A., & Crescitelli, S. (2017b). Stochastic foundations of undulatory transport phenomena: Generalized Poisson-Kac processes – Part iii extensions and applications to kinetic theory and transport. *Journal of Physics A: Mathematical and Theoretical*, 50(33), 335004. <https://doi.org/10.1088/1751-8121/aa79d6>
- Hassini, L., Azzouz, S., Peczkalski, R., & Belghith, A. (2007). Estimation of potato moisture diffusivity from convective drying kinetics with correction for shrinkage. *Journal of Food Engineering*, 79(1), 47–56.
- Helmroth, E., Varekamp, C., & Dekker, M. (2005). Stochastic modelling of migration from polyolefins. *Journal of the Science of Food and Agriculture*, 85(6), 909–916.
- Li, J., & Mason, D. (2002). Application of the discrete element modelling in air drying of particulate solids. *Drying Technology*, 20(2), 255–282.
- Liu, Q., & Bakker-Arkema, F. (1997). Stochastic modelling of grain drying. 2. Model development. *Journal of Agricultural Engineering Research*, 66(4), 275–280.
- Madiouli, J., Sghaier, J., Lecomte, D., & Sammouda, H. (2012). Determination of porosity change from shrinkage curves during drying of food material. *Food and Bioprocess Processing*, 90(1), 43–51.
- Mayor, L., & Sereno, A. (2004). Modelling shrinkage during convective drying of food materials: A review. *Journal of Food Engineering*, 61(3), 373–386.
- Nivelle, M. A., Beghin, A. S., Bosmans, G. M., & Delcour, J. A. (2018). Molecular dynamics of starch and water during bread making

- monitored with temperature-controlled time domain 1H NMR. *Food Research International*, 119, 675–682.
- Nowicki, S., Davis, H., & Scriven, L. (1992). Microscopic determination of transport parameters in drying porous-media. *Drying Technology*, 10(4), 925–946.
- Ortiz-Garcia-Carrasco, B., Yañez-Mota, E., Pacheco-Aguirre, F., Ruiz-Espinosa, H., Garcia-Alvarado, M., Cortés-Zavaleta, O., & Ruiz-López, I. (2015). Drying of shrinkable food products: Appraisal of deformation behavior and moisture diffusivity estimation under isotropic shrinkage. *Journal of Food Engineering*, 144, 138–147.
- Pacheco-Aguirre, F., Garcia-Alvarado, M., Corona-Jiménez, E., Ruiz-Espinosa, H., Cortés-Zavaleta, O., & Ruiz-López, I. (2015). Drying modeling in products undergoing simultaneous size reduction and shape change: Appraisal of deformation effect on water diffusivity. *Journal of Food Engineering*, 164, 30–39.
- Papanu, J., Soane (Soong), D., Bell, A., & Hess, D. (1989). Transport models for swelling and dissolution of thin polymer films. *Journal of Applied Polymer Science*, 38(5), 859–885.
- Pizzirusso, A., Brasiello, A., De Nicola, A., Marangoni, A., & Milano, G. (2015). Coarse-grained modelling of triglyceride crystallisation: A molecular insight into tripalmitin tristearin binary mixtures by molecular dynamics simulations. *Journal of Physics D: Applied Physics*, 48(49), 494004.
- Poschet, F., Geeraerd, A., Scheerlinck, N., Nicolai, B., & Van Impe, J. (2003). Monte Carlo analysis as a tool to incorporate variation on experimental data in predictive microbiology. *Food Microbiology*, 20(3), 285–295.
- Ramos, I., Brandao, T., & Silva, C. (2003). Structural changes during air drying of fruits and vegetables. *Food Science and Technology International*, 9(3), 201–206.
- Ruiz-López, I., Ruiz-Espinosa, H., Arellanes-Lozada, P., Bárcenas-Pozos, M., & Garcia-Alvarado, M. (2012). Analytical model for variable moisture diffusivity estimation and drying simulation of shrinkable food products. *Journal of Food Engineering*, 108(3), 427–435.
- Terrazas-Velarde, K., Peglow, M., & Tsotsas, E. (2009). Stochastic simulation of agglomerate formation in fluidized bed spray drying: A micro-scale approach. *Chemical Engineering Science*, 64(11), 2631–2643.
- Tu, Y.-O., & Ouano, A. C. (1977). Model for the kinematics of polymer dissolution. *IBM Journal of Research and Development*, 21(2), 131–142.
- Vu, H., & Tsotsas, E. (2018). Mass and heat transport models for analysis of the drying process in porous media: A review and numerical implementation. *International Journal of Chemical Engineering*, 2018, 1–13.
- Wang, N., & Brennan, J. (1995). Changes in structure, density and porosity of potato during dehydration. *Journal of Food Engineering*, 24(1), 61–76.
- Xu, P., Mujumdar, A. S., & Yu, B. (2008). Fractal theory on drying: A review. *Drying Technology*, 26(6), 640–650.
- Zielinska, M., & Markowski, M. (2010). Air drying characteristics and moisture diffusivity of carrots. *Chemical Engineering and Processing: Process Intensification*, 49(2), 212–218.

How to cite this article: Adrover A, Brasiello A. A moving boundary model for food isothermal drying and shrinkage: One-dimensional versus two-dimensional approaches. *J Food Process Eng*. 2019;42:e13178. <https://doi.org/10.1111/jfpe.13178>